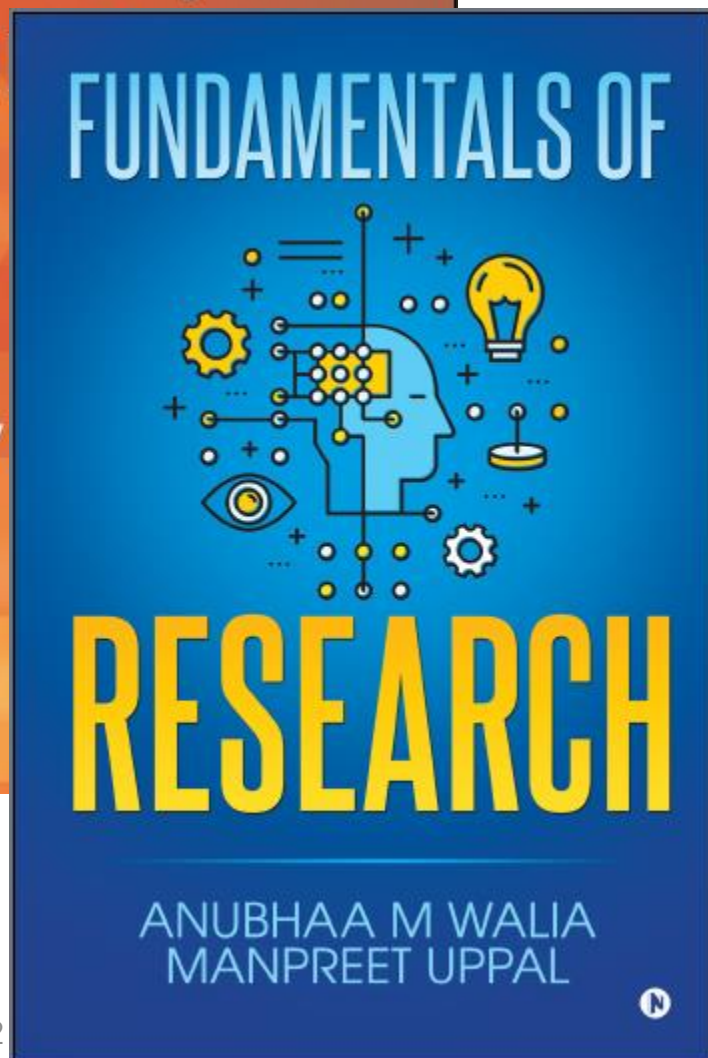


Research Fundamentals Unit - 1

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Prepared for BECE of NCIT



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- 1.2 Research Aim and Objectives
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What is Research?

- Research is creating some **new knowledge**
- Research is about **acquiring knowledge** and **developing understanding**, collecting facts and interpreting them to build up a picture of the world around us, and even within us.
- Research is a term used liberally for any kind of investigation that is intended **to uncover interesting or new facts**.
- Research Methods are the **tools and techniques** for doing research.
- Research designs are **plans and the procedures** for research that span the decisions from broad **assumptions** to detailed methods of **data collection and analysis**.

Research Aim and Objective

- To achieve skillfulness or to get different opinions by investigating a problem which is not clearly defined (**exploratory approach**)
- To establish the relationship with something which occurs or is related i.e., study concerning whether certain variables are associated. (**diagnostic approach**)

- Specific research objectives essentially focus on answering the **3W1H questions** by understanding problem constraints.
- These 3W1H are **What, Why, When, and How**.

- To **explore** a new idea.
- **Identifying factors** impacting XYZ
- To **gain insights** into the occurrence of a phenomenon.
- **To understand the characteristics** of an individual, situation, or event correctly & with accuracy.
- To identify the **frequencies of occurrence of events**.
- To establish a **cause and effect relationship**.
- To test the **relationship between two variables**.

Activities in Research

- ♦ **Categorize**
- ♦ **Describe**
- ♦ **Explain**
- ♦ **Evaluate**

- ♦ **Compare**
- ♦ **Correlate**
- ♦ **Predict**
- ♦ **Control**

- ♦ **Identify a problem**
- ♦ **Gather data**
- ♦ **Analyse the data**
- ♦ **Interpret the data**
- ♦ **Draw Conclusion**

Type of Research Design

Qualitative Research

Quantitative Research

Mixed Methods

Basic Research

Applied Research

Descriptive Research

Analytical Research

Applied Research

Fundamental Research

Conceptual Research

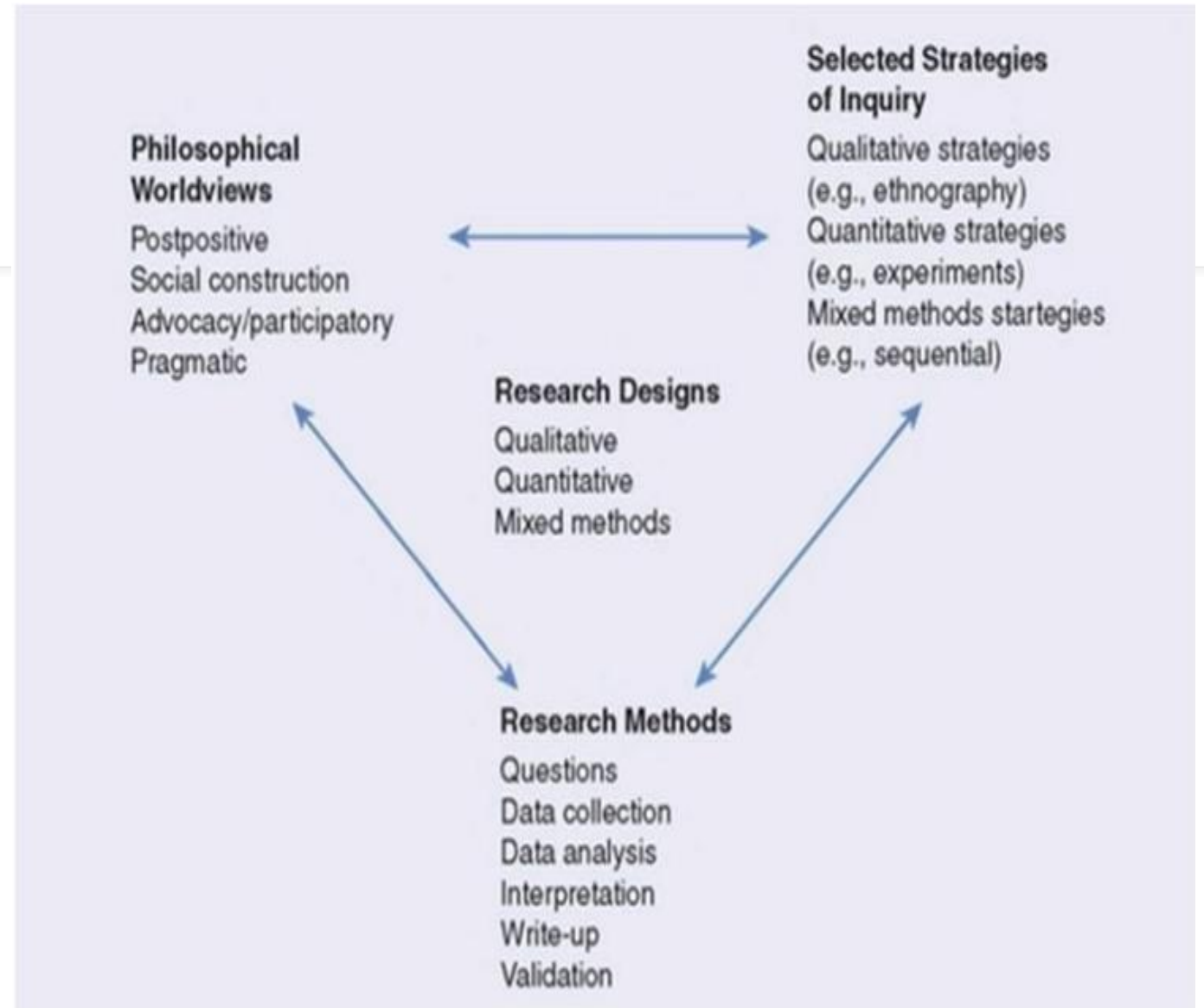
Empirical Research

Components involved in a Design

Intersection of Philosophy

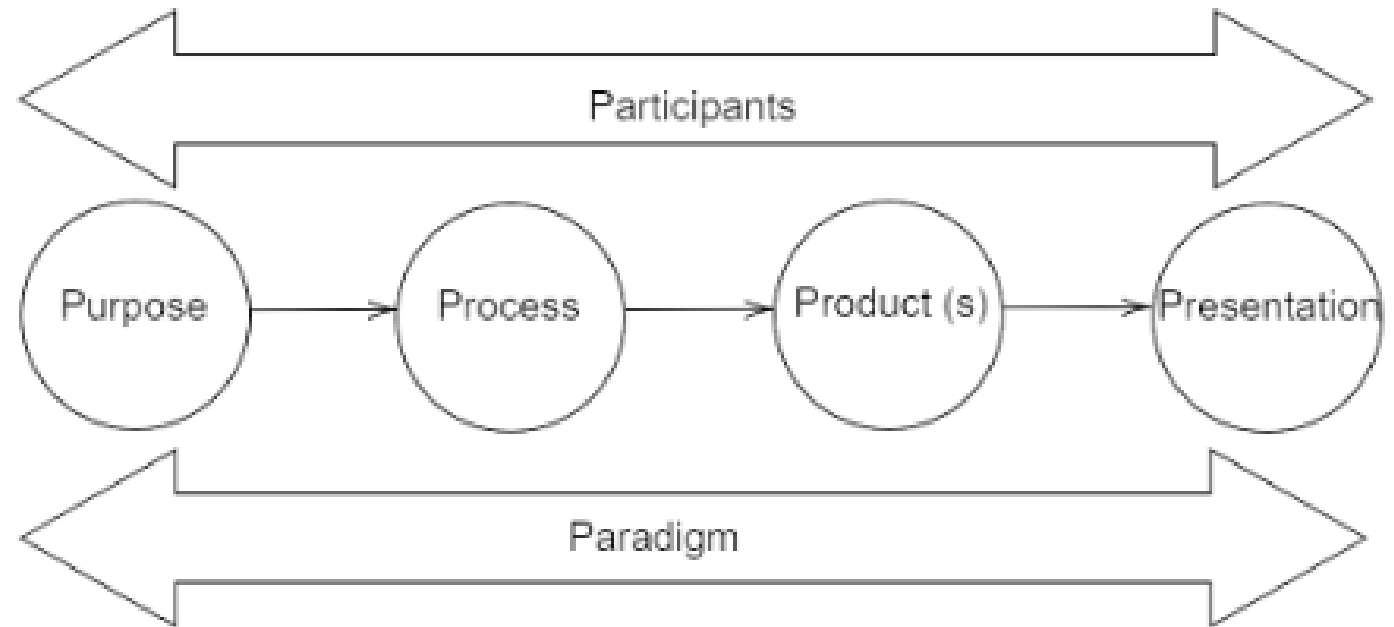
Strategies for Inquiry

Specific Methods



6Ps of Research

1. **Purpose**
2. **Products**
3. **Process**
4. **Participants**
5. **Paradigm**
6. **Presentation**



Purpose and Product of Research

- Purpose (the reason for doing it)
 - Add knowledge
 - Find out what happens
 - Solve a problem
 - Find the evidence to inform practice
 - Develop understating of the people and their world
 - Predict, plan and control
 - Contribute to people's well-being / personal needs
 - Test or disprove theory
 - Improve something
- Product (the outcome)
 - A new / improved product
 - A new / improved theory / model / perspective
 - Re-interpretation of an existing theory
 - Exploration of a topic / an area / a field
 - A critical analysis
 - Unanticipated outcomes

Research vs. Project

- **Research:**
- More theoretical
- Tests the effectiveness of a proposed intervention
- driven by critical inquiry, and leads to investigation, or theory testing, through established quantitative and/or qualitative methods.
- **Thesis:**
- It is an original contribution to knowledge resulting from the systematic study of a significant problem or issue.
- It advances our knowledge as a discipline.
- It makes an original and/or significant/or meaningful contribution to the knowledge base.



Research vs. Project

-
- **Project**
 - more **implementational**
 - leads to the **development of a program or a manual**, or a proposed intervention.
 - should **demonstrate excellence**, and may even be publishable. However, the project **may not carry the same level of originality** as the thesis
 - Types of project : (1) **creative** project, (2) **curriculum** project, (3) **investigative** or research project etc..

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Thank you



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